

Announcements

- Labs and Matlab
- Homework Policy
 - Hand in your own code and simulation
 - Collaboration is fine, but must explicitly state “I worked with Person X.”
- Monday 4:30 – 6pm section
 - TF: Katie Banks
 - Location: Pierce 307

Flu Pandemics

OPINION

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The Global Avian Threat

By BEDE A. MOORE,

Published: Friday, September 23, 2005

In recent months two significant events in U.S. history have served to government's failure and incapacity to fulfill the role for which they were created: the 9/11 attacks, and Hurricane Katrina. The latter, a civil strife, an increase in suicide bombings, and growing unrest prevalent in the Middle East. Hurricane Katrina ruined millions of lives, caused disastrous damage to property, and caused the deaths of thousands of people. It has caused virtually all Americans cause for thought about the abilities of the current administration to handle a crisis. Journalists quickly jumped on the bandwagon, clambering over one another to report on the human tragedy and loss of life. For weeks columns have lamented the government's slow response, and the suffering of fellow citizens—all with extremely good intentions. A more urgent response is necessary. Sadly, however, the war on terror and natural disasters have created demons that threaten the population.

Beyond the scope of the American continent there are far more concerns about global survival that took place in the last week. On Sept. 19, government officials in Indonesia shut down the Bogor Zoo when tests on an exotic bird

Posted on Sun, Feb. 7, 2010

Bird flu out of mind but still a threat

The Fatal Strain

On the Trail of Avian Flu and the Coming Pandemic

By Alan Sipress

Viking. 386 pp \$27.95

Reviewed by Huntly Collins

For most Americans, the threat of avian flu - "bird flu," in the vernacular - is likely a distant memory.

The highly pathogenic virus, which came to public attention in 1997 with the first human cases identified in Hong Kong, swept through the poultry industry in Southeast Asia and eventually spread to birds around the world.

But H5N1, as scientists call the virus, never gained much of a foothold in humans, mainly because it isn't easily transmitted from one person to another. Since 2003, there have been

Flu Pandemics

Swine Flu (H1N1) Vaccine

13 January 2011 Last updated at 11:14 ET



By THE NEW YORK TIMES

Sixty more deaths, but flu 'may be starting to plateau'

Overview

A vaccine for the H1N1 strain of [swine flu](#) became available in October 2009 as the country girded for a second wave of the pandemic that broke out across the country and much of the world in the spring. Despite shortages and confusion in the initial weeks of the vaccination effort, by December enough doses were available that officials expanded the effort beyond the high-risk groups initially targeted: pregnant women, people under 24 years old or caring for infants, people with high-risk medical conditions and health-care workers.

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By the end of the year, over 85 million doses were available. By February 2010, there was so much unused swine flu vaccine in the world that rich nations, including the United States, were trying to get rid of their surpluses. But logistical difficulties meant that the world's poorest countries — a few still facing the brunt of the pandemic — were receiving very little of it.

Medical experts looking back on the course of the epidemic -- although it may be far from over -- have credited luck and a series of rapid but conservative decisions by federal officials with limiting the virus's toll. In December, federal health officials estimated that 10,000 Americans had died since April of swine flu, far below the range of 30,000 to 90,000 predicted by a presidential panel.

Sixty-two people died with flu in the UK in the past week, bringing the total to 112 since October, figures reveal.

However, the Health Protection Agency (HPA) said the rise could be due to delayed reporting, with many of the deaths occurring before Christmas.

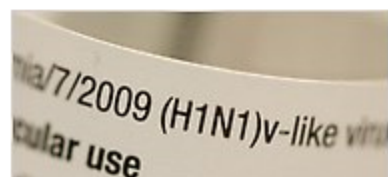
Most of those who died had swine flu. The majority were aged between 15 and 64, but nine children under 14 died.

An HPA spokesman said the number of people seeing their GP for flu-like illness was beginning to plateau.

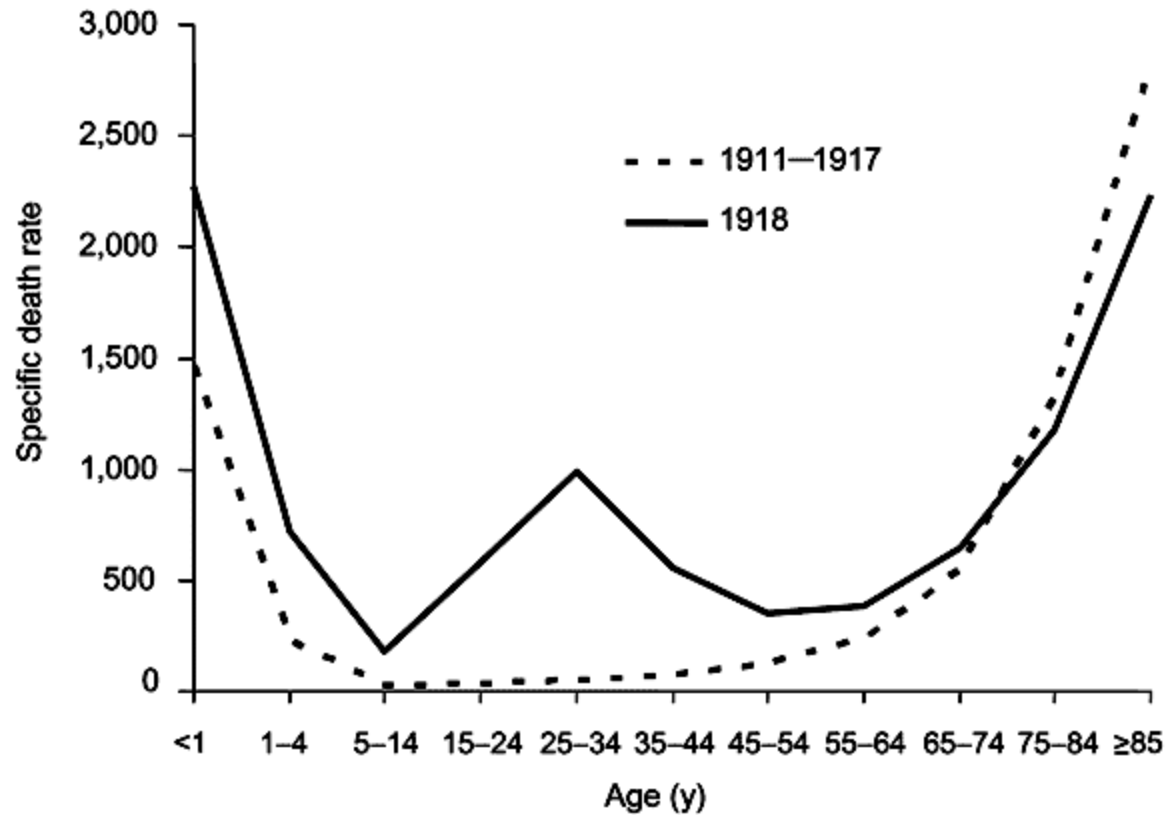
Of the 81 deaths where data is available, 63 people were in risk groups for flu.

And of the 47 for whom vaccine status is known, 40 had not had this season's flu jab.

~ 14,000 deaths worldwide



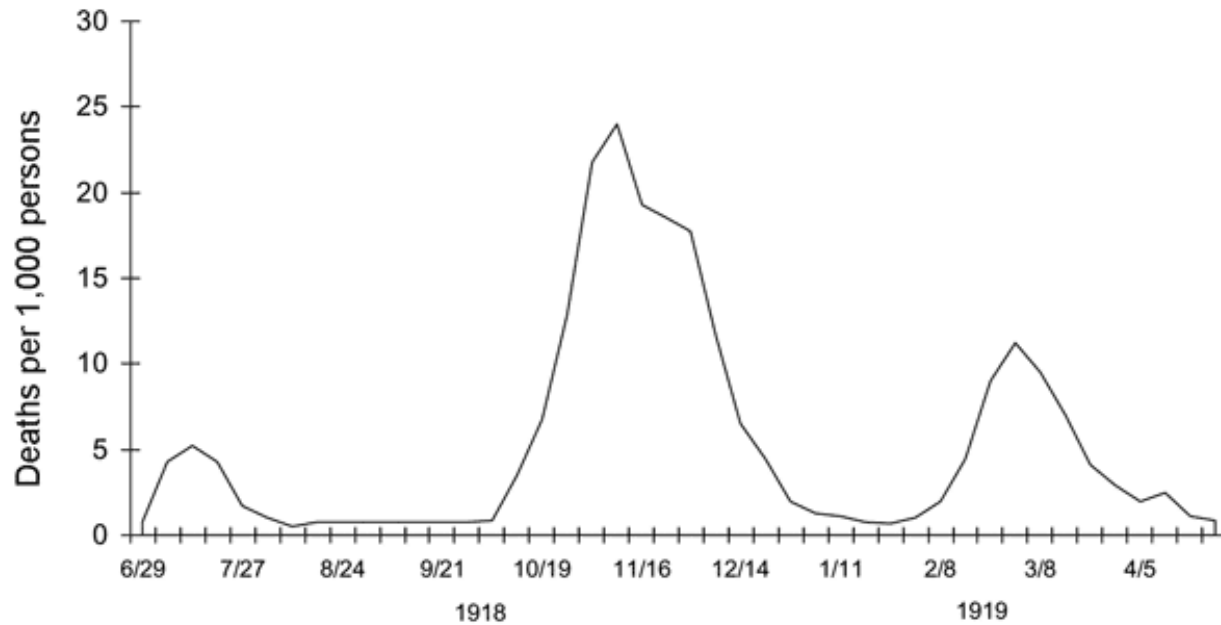
1918 Flu



Origin: China, Austria, Kansas?

Toll: 1/3 of world population infected
3-6% of world population died

Mortality in UK: waves of the flu



Questions

1. Why are some flu's pandemics and some not? Why are pandemics so sudden?
2. Why do flu's end? Perhaps all people who can get it either recover or are "removed" from the population?
3. How can we anticipate a flu pandemic, in *early* stages?
4. Vaccination strategies—how many people should we vaccinate, etc?
5. Suppose someone in this room had bird flu or H1N1. How worried should you be that you'll be infected?

A Contribution to the Mathematical Theory of Epidemics.

By W. O. KERMAK and A. G. MCKENDRICK.

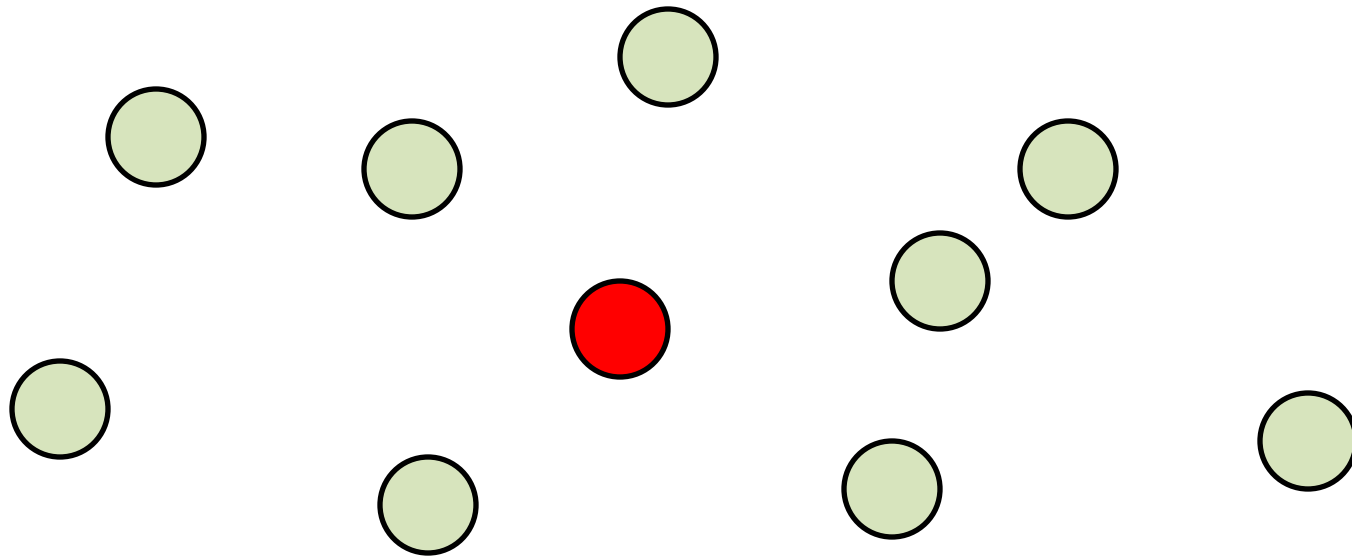
(Communicated by Sir Gilbert Walker, F.R.S.—Received May 13, 1927.)

(From the Laboratory of the Royal College of Physicians, Edinburgh.)

Introduction.

(1) One of the most striking features in the study of epidemics is the difficulty of finding a causal factor which appears to be adequate to account for the magnitude of the frequent epidemics of disease which visit almost every population. It was with a view to obtaining more insight regarding the effects of the various factors which govern the spread of contagious epidemics that the present investigation was undertaken.

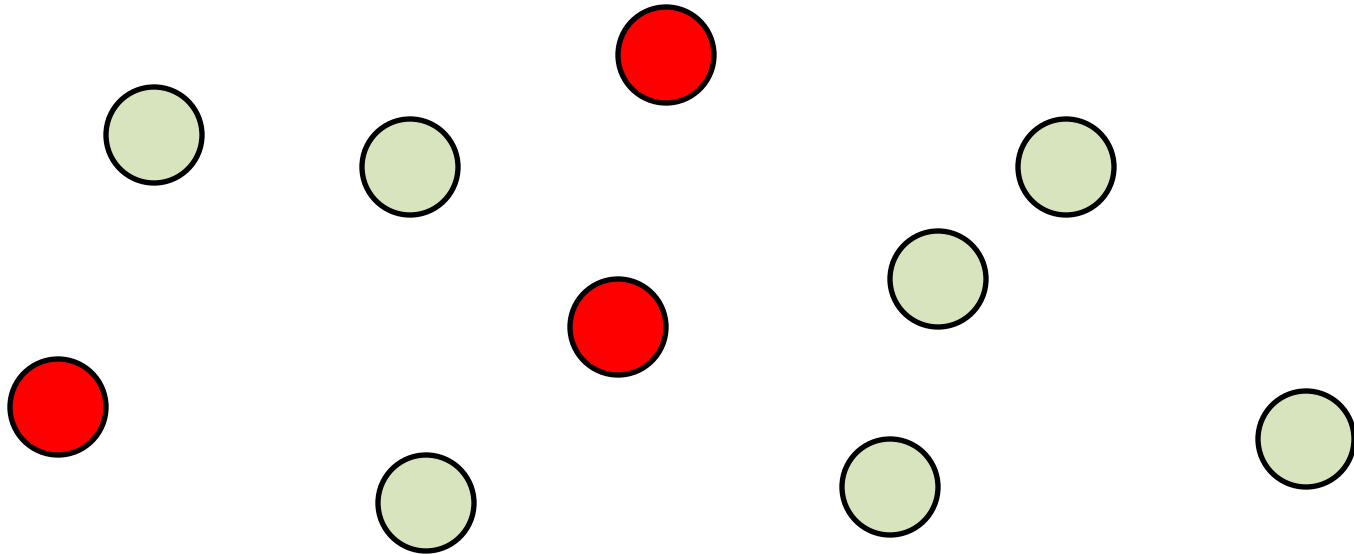
General Idea



Three populations:

Susceptible	$S(t)$	
Infected	$I(t)$	
Removed	$R(t)$	

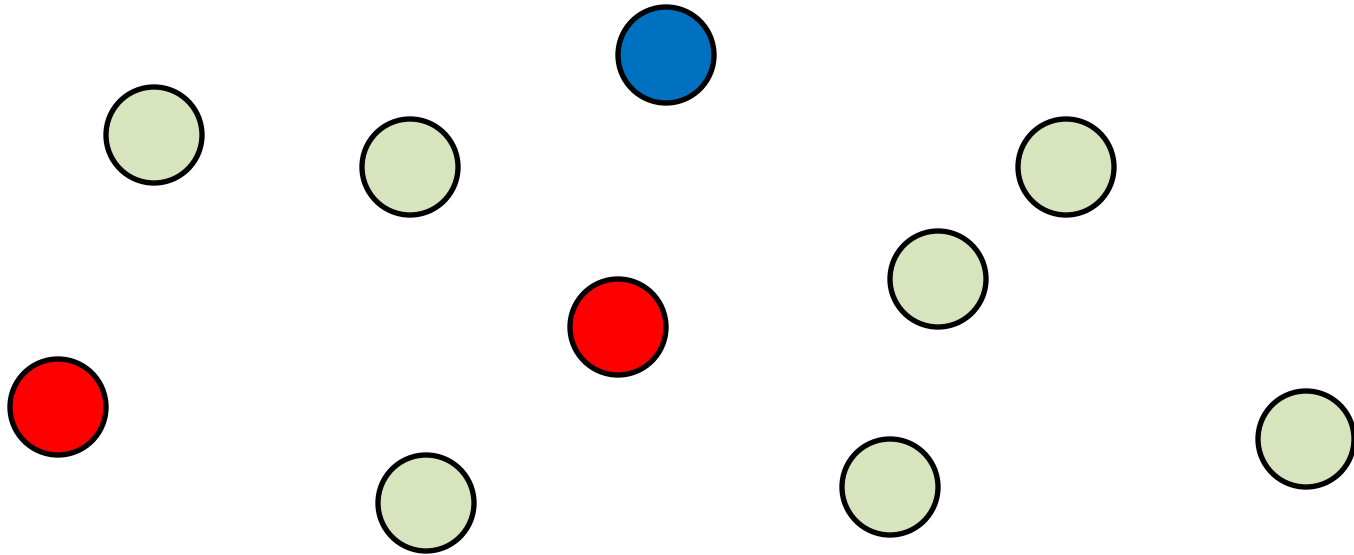
General Idea



Three populations:

Susceptible	$S(t)$	
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Removed	$R(t)$	

General Idea



Three populations:

Susceptible	$S(t)$	
Infected	$I(t)$	
Removed	$R(t)$	

How do we turn this idea into equations?

x_t : Number of people not yet infected 

y_t : Number of people infected 

z_t : Number of people removed (recovered/dead) 

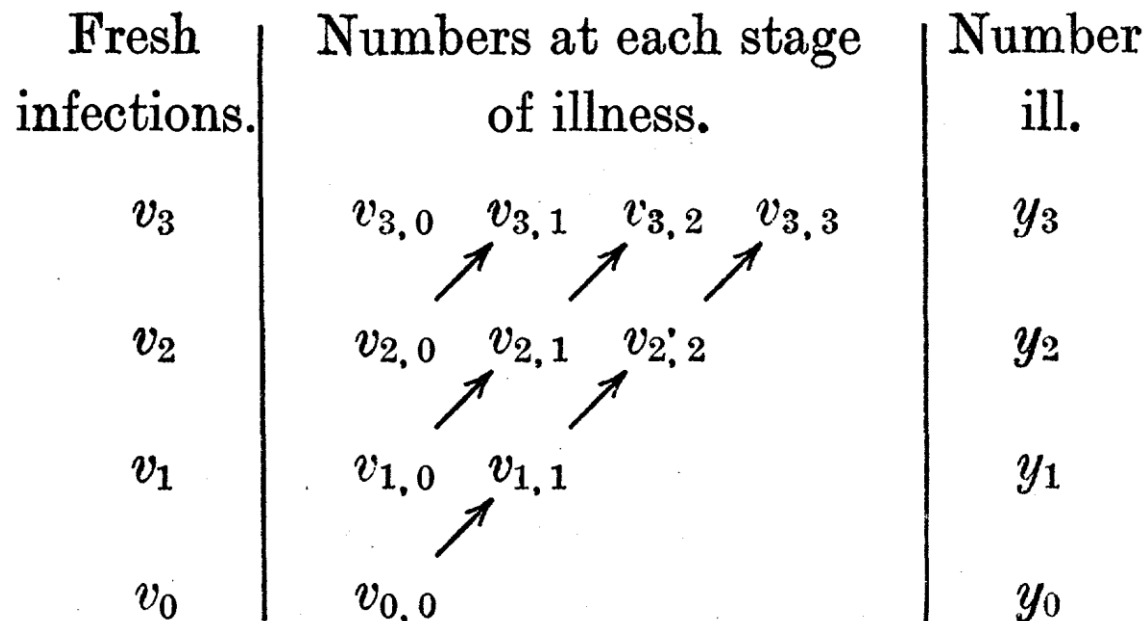
$v_t = x_t - x_{t+1}$ number of people infected at time t

$y_t = \sum_{\theta=0}^t v_{t,\theta} = v_{t,0} + v_{t,1} + \dots + v_{t,t}$

 number of people sick at time t
who were infected at time θ

$x_t + y_t + z_t = N$ size of population

How do we turn this idea into equations?



$v_{t,\theta}$: number of people sick at time t
who were infected at time θ

y_t : total number of people sick at time t

v_t : number of people who get sick at time t

$$x_t + y_t + z_t = N$$

$$v_t = -\frac{dx_t}{dt}$$

$$\frac{dx_t}{dt} = -x_t \left[\int_0^t A_\theta v_{t-\theta} d\theta + A_t y_0 \right]$$

$$\frac{dz_t}{dt} = \int_0^t C_\theta v_{t-\theta} d\theta + C_t y_0$$

$$y_t = \int_0^t B_\theta v_{t-\theta} d\theta + B_t y_0$$

A_t : Related to rates of infection and removal

B_t C_t : Related to rate of removal

The paper then attempts to “solve” these equations

Pre-digital computers!

Simulations could not give insight into solutions

Manages to derive a formula for the size of an epidemic

$$-\log \frac{1-p}{1-\frac{y_0}{N}} = ANp$$

p : proportion of population infected during an epidemic

We will do this for a simpler model

Simpler model: constant rates

B. *Constant Rates.*

Much insight can be obtained as to the process by which epidemics in limited populations run their peculiar courses, and end in final extinction, from the consideration of the special case in which ϕ and ψ are constants κ and l respectively.

In this case the equations are

$$\frac{dx}{dt} = -\kappa xy$$

$$\frac{dy}{dt} = \kappa xy - ly$$

$$\frac{dz}{dt} = ly$$

and as before $x + y + z = N$.

Modern Language (SIR)

$S(t)$ Susceptible 

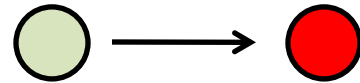
$I(t)$ Infected 

$R(t)$ Removed 

$$\frac{dS}{dt} = -\beta IS$$

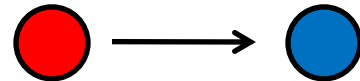
Infection rate





$$\frac{dI}{dt} = \beta IS - \gamma I$$

Recovery/death rate

$$\frac{dR}{dt} = \gamma I$$

An AMAZING Result

$$\frac{dI}{dt} = \beta IS - \gamma I = I(\beta S - \gamma)$$

$$\frac{dI}{dt} > 0 \quad \Rightarrow \quad \boxed{S > \frac{\gamma}{\beta}} \quad \text{Epidemic spreads!}$$

β : rate of infection of susceptible people

γ : rate of recovery/death of infected people

For example ...

γ : rate of recovery/death of infected people

i.e. two weeks to recover from the flu

β : rate of infection of susceptible people

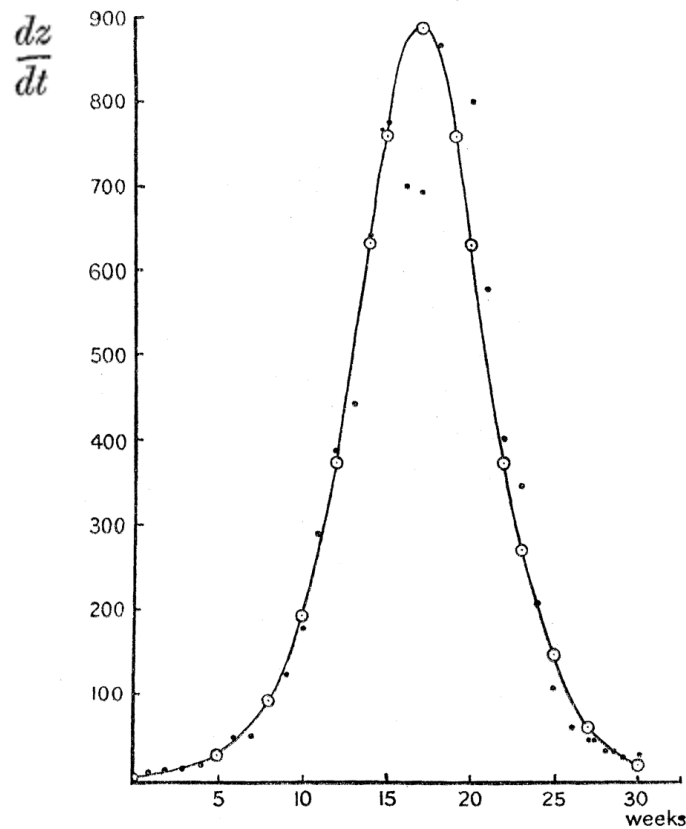
i.e. in one hour $1/N$ will get sick if an infected person is in the room

$$S > \frac{\gamma}{\beta} = \frac{1 \text{ recovery}/(2*7*24 \text{ hours})}{(1/N) \text{ infection/hour}} = N/336$$

The model therefore predicts that critical population size depends very sensitively on infection rate, as well as the number of days you're infectious!

$$N_{cr} = \frac{\gamma}{\beta}$$

These are quite variable, as we know ...



The accompanying chart is based upon figures of deaths from plague in the island of Bombay over the period December 17, 1905, to July 21, 1906. The ordinate represents the number of deaths per week, and the abscissa denotes the time in weeks. As at least 80 to 90 per cent. of the cases reported terminate fatally, the ordinate may be taken as approximately representing dz/dt as a function of t . The calculated curve is drawn from the formula

$$\frac{dz}{dt} = 890 \operatorname{sech}^2 (0.2t - 3.4).$$

We are, in fact, assuming that plague in man is a reflection of plague in rats, and that with respect to the rat (1) the uninfected population was uniformly susceptible ; (2) that all susceptible rats in the island had an equal chance of being infected ; (3) that the infectivity, recovery, and death rates were of constant value throughout the course of sickness of each rat ; (4) that all cases ended fatally or became immune ; and (5) that the flea population was so large that the condition approximated to one of contact infection. None of these assumptions are strictly fulfilled and consequently the numerical equation can only be a very rough approximation. A close fit is not to be expected, and deductions as to the actual values of the various constants should not be drawn. It may be said, however, that the calculated curve, which implies that the rates did not vary during the period of the epidemic, conforms roughly to the observed figures.

Conclusions

2. In general a threshold density of population is found to exist, which depends upon the infectivity, recovery and death rates peculiar to the epidemic. No epidemic can occur if the population density is below this threshold value.

3. Small increases of the infectivity rate may lead to large epidemics ; also, if the population density slightly exceeds its threshold value the effect of an epidemic will be to reduce the density as far below the threshold value as initially it was above it.

4. An epidemic, in general, comes to an end, before the susceptible population has been exhausted.

5. Similar results are indicated for the case in which transmission is through an intermediate host.



Detecting influenza epidemics using search engine query data

Jeremy Ginsberg¹, Matthew H. Mohebbi¹, Rajan S. Patel¹, Lynnette Brammer², Mark S. Smolinski¹ & Larry Brilliant¹

¹Google Inc. ²Centers for Disease Control and Prevention

We sought to develop a simple model which estimates the probability that a random physician visit in a particular region is related to an influenza-like illness (ILI); this is equivalent to the percentage of ILI-related physician visits. A single explanatory variable was used: the probability that a random search query submitted from the same region is ILI-related, as determined by an automated method described below. We fit a linear model using the log-odds of an ILI physician visit and the log-odds of an ILI-related search query:

$$\text{logit}(P) = \beta_0 + \beta_1 \times \text{logit}(Q) + \varepsilon$$

where P is the percentage of ILI physician visits, Q is the ILI-related query fraction, β_0 is the intercept, β_1 is the multiplicative coefficient, and ε is the error term. $\text{logit}(P)$ is the natural log of $P/(1-P)$.

Publicly available historical data from the CDC's U.S. Influenza Sentinel Provider Surveillance Network¹² was used to help build our models. For each of the nine surveillance regions of the United States, CDC reported the average percentage of all outpatient visits to sentinel providers that were ILI-related on a weekly basis. No data were provided for weeks outside of the annual influenza season, and we excluded such dates from model fitting, though our model was used to generate unvalidated ILI estimates for these weeks.

Search Query Topic	Top 45 Queries		Next 55 Queries	
	N	Weighted	N	Weighted
Influenza Complication	11	18.15	5	3.40
Cold/Flu Remedy	8	5.05	6	5.03
General Influenza Symptoms	5	2.60	1	0.07
Term for Influenza	4	3.74	6	0.30
Specific Influenza Symptom	4	2.54	6	3.74
Symptoms of an Influenza Complication	4	2.21	2	0.92
Antibiotic Medication	3	6.23	3	3.17
General Influenza Remedies	2	0.18	1	0.32
Symptoms of a Related Disease	2	1.66	2	0.77
Antiviral Medication	1	0.39	1	0.74
Related Disease	1	6.66	3	3.77
Unrelated to Influenza	0	0.00	19	28.37
	45	49.40	55	50.60

Table 1: Topics found in search queries which were found to be most correlated with CDC ILI data. The top 45 queries were used in our final model; the next 55 queries are presented for comparison purposes. The number of queries in each topic is indicated, as well as query volume-weighted counts, reflecting the relative frequency of queries in each topic.

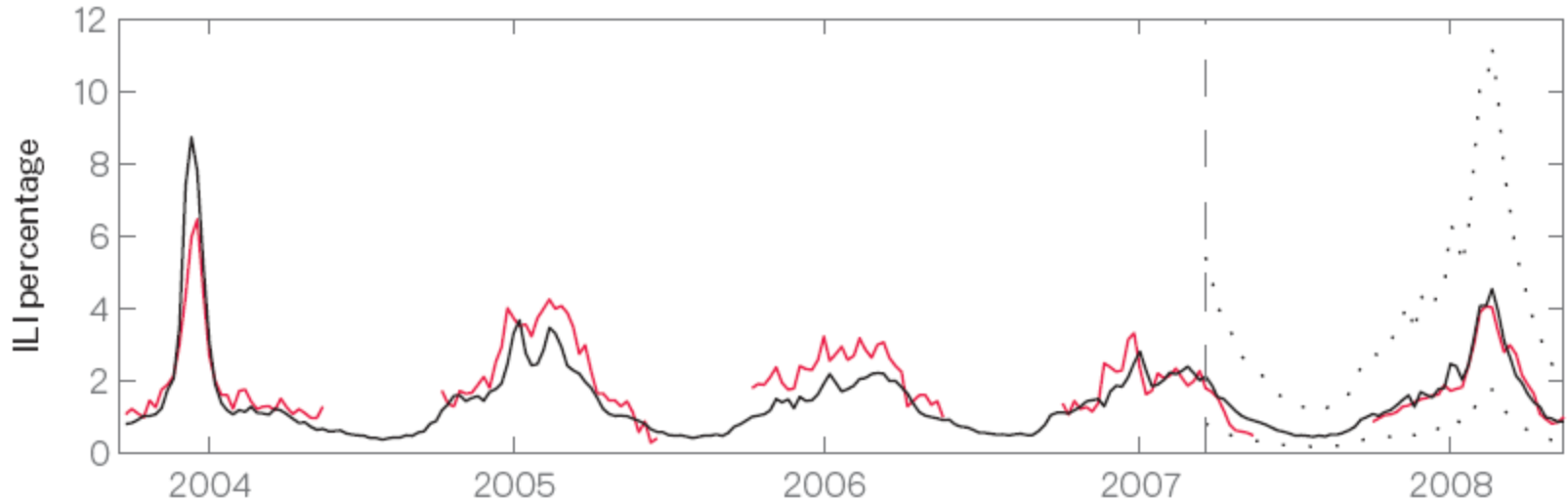


Figure 2: A comparison of model estimates for the Mid-Atlantic Region (black) against CDC-reported ILI percentages (red), including points over which the model was fit and validated. A correlation of 0.85 was obtained over 128 points from this region to which the model was fit, while a correlation of 0.96 was obtained over 42 validation points. 95% prediction intervals are indicated.

[Google.org home](#)

Flu Trends

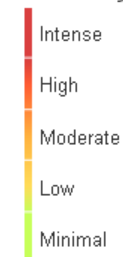
Home

Select country/region 

[How does this work?](#)

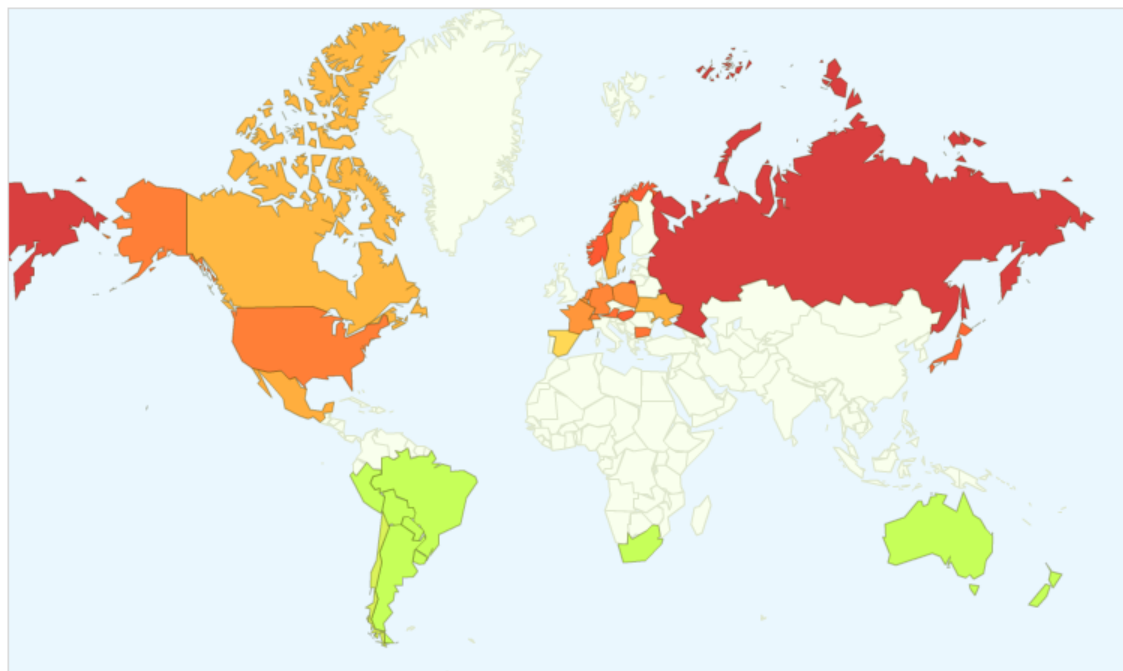
[FAQ](#)

Flu activity



Explore flu trends around the world

We've found that certain search terms are good indicators of flu activity. Google Flu Trends uses aggregated Google search data to estimate flu activity. [Learn more >](#)



[Download world flu activity data](#) - [Animated flu trends for Google Earth](#) - [Compare flu trends across regions in Public Data Explorer](#)

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Flu Trends

[Home](#)

United States

National

[Download data](#)

[How does this work?](#)

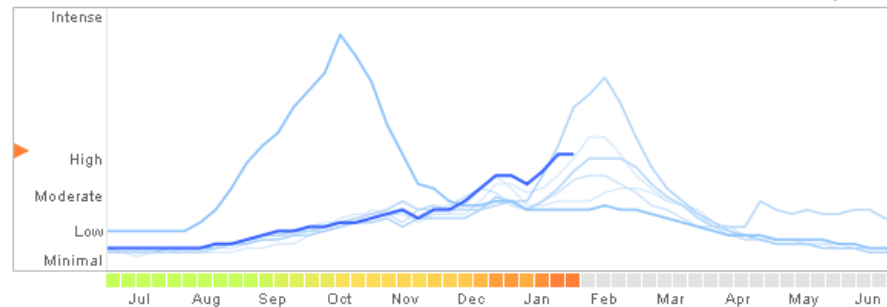
[FAQ](#)

Explore flu trends - United States

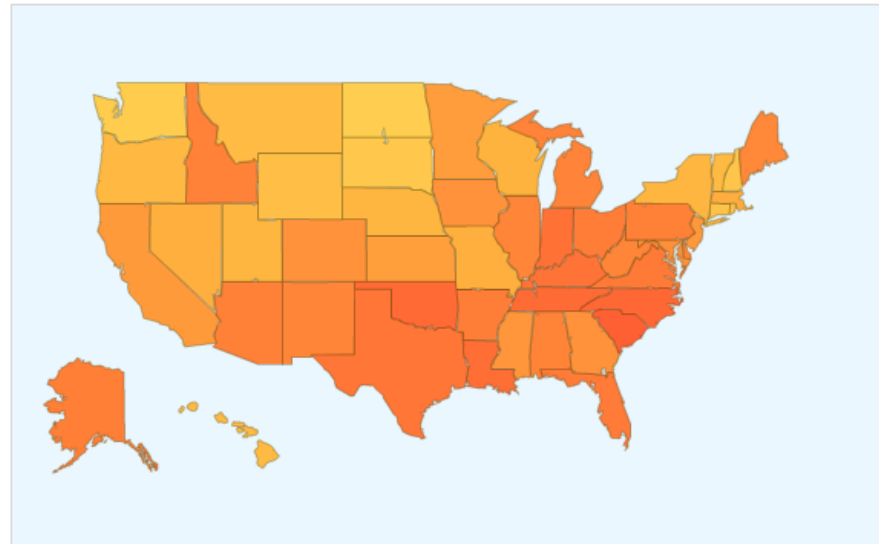
We've found that certain search terms are good indicators of flu activity. Google Flu Trends uses aggregated Google search data to estimate flu activity. [Learn more »](#)

National

2010-2011 Past years



States | [Cities](#) (Experimental)



Estimates were made using a model that proved accurate when compared to historic official flu activity data. Data current through February 5, 2011.

Fight influenza

CDC urges you to take these steps to protect yourself and others from the flu:

1. Get vaccinated against flu – it's your best defense.
2. Cover your cough, wash hands often.
3. Take antiviral drugs if your doctor recommends them.

 [Centers for Disease Control and Prevention](#)

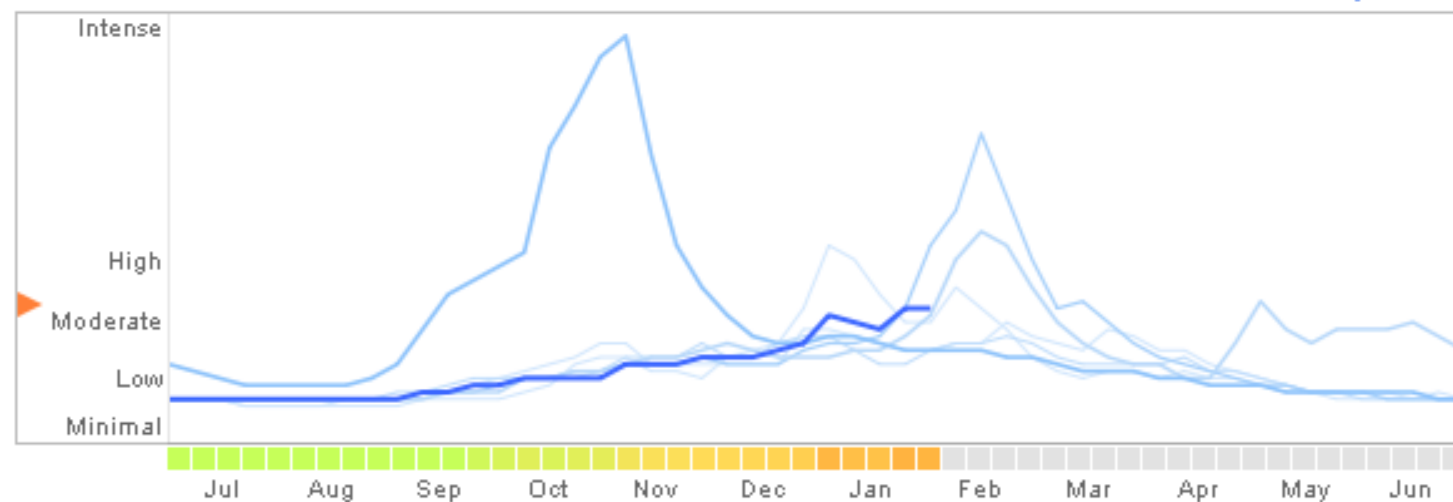
Animated Flu Trends in Google Earth

[Download and explore](#) Flu Trends data in Google Earth. Need Google Earth?

[Download it here.](#)

[United States](#) > Boston, MA

● 2010-2011 ● [Past years](#) ▼



Lab Questions

1. Solve SIR model numerically

Experiment with different parameters and explore size of epidemic

2. Think about the role of vaccination

What fraction of a population needs to be vaccinated

3. Download data from Google flu

Play around with SIR model parameters and try to fit the data.

4. Questions to think about

Geographical spread of disease? Do curves make sense in light of relative parameters in different states, countries?